

NETWORKS AND COMPLEXITY

Solution 20-1

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 20.1: Multi-step adjacency [Lvl. 1]

Let λ_n be the eigenvalues of $\mathbf{A}$. Show that the eigenvalues of $\mathbf{A}^k$ are λ_n^k .

Solution

We start from the eigenvector equation

$$\mathbf{A}\mathbf{v}_n = \lambda_n \mathbf{v}_n. \quad (1)$$

Now we can show that

$$\mathbf{A}^k \mathbf{v}_n = \mathbf{A}^{k-1} \mathbf{A} \mathbf{v}_n \quad (2)$$

$$= \mathbf{A}^{k-1} \lambda_n \mathbf{v}_n \quad (3)$$

$$= \mathbf{A}^{k-2} \lambda_n \mathbf{A} \mathbf{v}_n \quad (4)$$

$$= \mathbf{A}^{k-2} \lambda_n^2 \mathbf{v}_n \quad (5)$$

$$= \mathbf{A}^{k-m} \lambda_n^m \mathbf{v}_n \quad (6)$$

$$= \mathbf{A}^0 \lambda_n^k \mathbf{v}_n \quad (7)$$

$$= \lambda_n^k \mathbf{v}_n \quad (8)$$

In summary

$$(\mathbf{A}^k) \mathbf{v}_n = (\lambda_n^k) \mathbf{v}_n \quad (9)$$

which shows that $\mathbf{v}_n$ is an eigenvector of $\mathbf{A}^k$ with eigenvalue λ_n^k .